

Network security incident analysis using NIST CSF

1. Summary

The organization experienced a denial-of-service (DoS) attack that disrupted network operations for approximately two hours. The attack was caused by a flood of incoming ICMP packets that overwhelmed network resources and prevented normal traffic from accessing services. The cybersecurity team identified that the attack was made possible by an improperly configured firewall. To mitigate the incident, the team blocked malicious traffic, restored critical services, and implemented additional security controls.

A) IDENTIFY

The organization was targeted by a malicious actor who launched an ICMP flood attack against the internal network. As a result, network services became unavailable and employees were unable to access critical resources. The investigation revealed that an unconfigured firewall allowed excessive ICMP traffic into the network, creating a vulnerability that enabled the attack.

B) PROTECT

To reduce the likelihood of future attacks, the cybersecurity team implemented new firewall rules to limit the rate of incoming ICMP packets. An IDS/IPS solution was also deployed to filter suspicious traffic before it could reach critical systems. These controls provide additional protection against denial-of-services attacks and improve the organization's overall security posture.

C) DETECT

To reduce the likelihood of future attacks, the cybersecurity team implemented new firewall rules to limit the rate of incoming ICMP packets. An IDS/IPS solution was also deployed to filter suspicious traffic before it could reach critical systems.

D) RESPOND

During the incident, the cybersecurity team blocked incoming malicious ICMP traffic and isolated affected services to prevent further disruption. Critical network resources were prioritized and restored as quickly as possible. The team also analyzed network logs to determine the attack vector and document the incident for future reference.

E) RECOVER

Recovery efforts focused on restoring normal network operations and ensuring that critical services remained available. Once the malicious traffic was mitigated, essential systems were brought back online first, followed by non-critical services. Additional firewall followed by non-critical services. Additional firewall protections, monitoring capabilities, and incident response procedures were implemented to improve resilience against future attacks.

F) LESSON LEARNED

The incident demonstrated the importance of proper firewall configuration, continuous network monitoring, and layered security controls. Organizations should regularly review firewall rules, monitor network traffic, and maintain effective incident response procedures to minimize the impact of future denial-of-services attack.